KARNATAKA STATE POLICE

Crime Intelligence & Analytics Portal

Comprehensive Codebase Analysis, System Flows, Core Features, Real-world Application, and Production-grade Upgrade Roadmap

Report Type: Technical Architecture Brief	**Target Audience:** Command Staff & Developers
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1. Executive Summary

The Karnataka State Police (KSP) Crime Intelligence & Analytics Portal is a multi-role tactical intelligence application. It integrates spatial mapping, syndicate networks, recidivism risk assessment, and financial tracking with a state-of-the-art conversational AI assistant that operates in dual modes: a cloud-based option leveraging Google Gemini 2.5 Flash, and a 100% offline option using local Ollama services (Gemma2/Llama3). The codebase is structured as a full-stack web application built with a React 19 + TypeScript frontend and a Node.js + Express backend that emulates a Zoho Catalyst serverless environment.

The system enforces strict role-based interfaces tailored for four distinct user archetypes: Sub-Inspectors (Investigators), District Analysts (Analysts), Assistant Commissioners (Supervisors), and the Director General of Police (Policymakers). Each dashboard serves specific tactical and operational goals, connected through a relational datastore and audited using cryptographic tamper-evident hash chaining.

ANALYSIS SCOPE: This analysis details the prototype's source code, evaluates interactive operational features, outlines real-world deployment cases, and details the production-grade roadmap to migrate from the emulated environment to live Zoho Catalyst cloud services.

2. System Architecture & Codebase Design

The codebase is divided into clear functional blocks that represent a modern microservices architecture, built to run on the Zoho Catalyst serverless environment. The files are distributed as follows:

- **Frontend Client (/client)**: A React 19 single-page application configured with Vite and TypeScript. Styling is handled via Tailwind CSS v4. Key visualization dependencies include *Leaflet* (geographic maps), *vis-network* (force-directed syndicate and financial flow graphs), and *Recharts* (macro charts).
- **Backend Functions (/functions)**: Emulates Catalyst's Serverless Functions. Subfolders represent standalone microservices. The entry point is server.js, which acts as an API gateway routing requests to modules like chat-handler, tool-router, risk-scoring, financial-analysis, cctns-client, pdf-export, and audit-log.
- **Relational Datastore (/datastore)**: Emulates Catalyst's relational data store using SQLite (ksp_crime.db). A SQL schema file (schema.sql) defines the tables and indexes. A Python seed script (generate_synthetic_data.py) generates realistic mock data including FIR reports, suspect profiles, socio-demographic indicators, and financial transactions.
- **Circuits & Configuration (/circuits, /ml)**: Emulates Catalyst Circuits for serverless workflow orchestration, and contains offline configurations for RAG indices and AutoML models.

Prototype to Production Catalyst Mapping

Prototype Module	Emulated Service	Production Equivalent	Core Role in Production
functions/chat-handler	Express.js API Node	Catalyst AppSail / Functions	Entry point for conversational engine APIs
functions/rag-query	SQLite RAG & Embeddings	Catalyst QuickML (RAG)	Vectorized search on FIR summaries
functions/risk-scoring	Algorithmic modifier engine	Catalyst Zia AutoML	Tabular recidivism risk scoring models
functions/translate	Police English-Kannada Dict	Catalyst Zia Translation	Dual-translation API for vernacular support
functions/pdf-export	HTML string parser	Catalyst SmartBrowz	Official KSP letterhead PDF compiler
datastore/ksp_crime.db	Local SQLite Datastore	Catalyst Data Store	Secure relational crime intelligence tables
circuits/query-orchestration	Express handler sequence	Catalyst Circuits	State-machine based query workflow

3. Core Features & Role-based Flow Analysis

The application is structured into four primary workflows matching police roles, plus a system-wide SmartBrowz control interface:

A. SI (Investigator) Dashboard: Active Case Solving

Designed for frontline Sub-Inspectors (SIs), this screen integrates case file tracking and suspect dossiers. It queries the datastore to show the current FIR details, mapped legal sections (using the IPC-to-BNS mapping), victim profiles, and case narratives. In addition, the SI can view the Accused Profile, which displays the suspect's demographics, prior convictions, and a Zia-calculated Recidivism Risk score. The SI can inspect the Digital Evidence vault where cryptographic file hashes are verified, showing a tamper-evident audit ledger, and view witness statements. A conversational AI assistant panel sits side-by-side to assist in natural language query analysis.

B. DA (Analyst) Dashboard: Spatial and Network Intelligence

Designed for District Analysts (DAs), this screen offers district-wide and state-wide visual maps. It features a Geographic Hotspot Map (built with Leaflet) rendering pins at crime coordinates. DAs can filter incidents by district and crime type, revealing hotspots. In the same view, the Syndicate Linkage Network (built with vis-network) renders force-directed graphs showing ties between repeat offenders, gang associations, and matching modus operandi. The DA also analyzes socio-demographic charts representing crime statistics alongside district unemployment and literacy rates, helping formulate early warnings.

C. ACP (Supervisor) Dashboard: Governance & Compliance

Designed for Assistant Commissioners of Police (ACPs), this dashboard provides operational oversight and audit controls. It displays a live Microservice Health Monitor checking connections to the database and Zia ML services. The centerpiece is the Live Audit Logs table. Every database query, PDF export, or system action written by investigators and analysts is recorded in an audit ledger. Each log entry is cryptographically linked to the previous log using a chain-hashing mechanism (SHA256), rendering the audit logs tamper-evident. The ACP can resolve compliance flags via override modals.

D. DGP (Policymaker) Executive Brief: Macro Planning

Designed for the Director General of Police (DGP), this dashboard is high-density and strategic. It displays statewide KPIs (active cases, critical hotspots, monitored districts) alongside crime trend timelines (Recharts) and socio-demographic crime charts. The DGP uses the interactive Intelligence Brief panel to customize key findings and recommendations, generating an official KSP-branded PDF report compiling the narrative automatically.

E. SmartBrowz Dashboard: Headless Browser & Automated Verification Logs

This administrative dashboard tracks headless browser automation, PDF exports, and dataverse synchronization logs. It includes a terminal simulator that visualizes script executions (e.g. bypass audits, template compilations) and renders charts showing performance latency and transaction volume over time.

Prototype Code Reference: Row-Level Security Emulation

In the prototype, Row-Level Security (RLS) is emulated in Express middleware. The server reads the JWT claims representing the user's role and district, and injects filters into SQL query parameters:

```
// functions/auth-middleware/rls-filter.js
module.exports = (req, res, next) => {
  const { role, district } = req.user; // Decoded JWT claims
  req.rlsFilter = {};

  if (role === 'Investigator') {
    // Restricted to user's assigned district
    req.rlsFilter.queryModifier = "WHERE district = ?";
    req.rlsFilter.params = [district];
  } else if (role === 'Analyst') {
    // Restricted to user's district but wider access
    req.rlsFilter.queryModifier = "WHERE district = ?";
    req.rlsFilter.params = [district];
  } else {
    // ACP (Supervisor) and DGP (Policymaker) have statewide access
    req.rlsFilter.queryModifier = "";
    req.rlsFilter.params = [];
  }
  next();
};
```

4. Phase 6 Interactive Upgrades & Operational Flows

Four critical interactive subsystems have been implemented to simulate real-world operations in the KSP portal:

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1. Interactive Recidivism Risk Calculator

Within the Investigator Dashboard, the suspect profile features dynamic threat toggles: *Active Arrest Warrant*, *Weapon Association*, *Hawala Money Linkage*, and *Habitual Offender Status*. Changing these values and clicking "Recalculate Profile" submits a modifier payload back to the chat assistant. The model dynamically recalculates the recidivism threat index (raising or lowering the score) and generates a fresh intelligence narrative explaining the modified score (e.g. highlighting how hawala connections signal syndicate ties).

2. CCTNS Sync Terminal Simulator

To simulate real-time data ingestion from the national CCTNS registry, the app header displays a live connection status. Clicking the sync button triggers a terminal emulator modal showing detailed operational logs in real-time (establishing secure handshake, query routing, parsing JSON schemas, and recalculating models) with progressive increments, concluding with the number of imported criminal files written to the SQLite datastore.

3. Supervisor Audit & Compliance Overrides

When an investigator triggers an anomalous query or access, it raises a predictive intelligence alert flag on the Supervisor Panel. The ACP can review the alert and override/resolve it by inputting an official compliance justification. Submitting this form writes the justification directly to the SQLite datastore, creating an immutable compliance audit ledger entry that is cryptographically chained, securing the department against unauthorized audits.

4. Lateral Map Drawer

When the Analyst clicks on any Leaflet map pin, the system handles the onIncidentSelect event by opening a sliding panel from the right. This panel displays the FIR details, modus operandi, suspect dossiers, and predictive warnings without cluttering the main map, creating a seamless UX for tactical control.

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5. Real-World Applications & Use Cases

This platform is designed to solve critical operational bottlenecks faced by state law enforcement agencies. The following scenarios illustrate how the portal's features translate to the field:

Scenario A: Combating Organized Cyber-Financial Syndicates

Operation: An investigator (SI) is investigating a phishing complaint where a victim lost Rs. 15 Lakhs. The investigator enters the query: *'Trace the money flow for case FIR-2026-003'* into the AI Assistant. The backend retrieves the financial transaction table and returns a visual Money Flow graph. The investigator immediately notices a red-flagged hawala node transferring money to an account associated with a suspect named 'Rupa Naik'. The investigator clicks on Rupa's profile, toggles the 'Hawala Money Linkage' modifier, and recalculates the risk. The AI assistant outputs a threat warning indicating Rupa Naik has a high risk profile and has ties to an active syndicate operating in Mysuru and Hubballi. The SI pins this to the Collaborative Desk so the District Analyst (DA) is notified.

Scenario B: Real-time Incident Geofencing & Dispatch Control

Operation: A suspect on the watch list pings a cell tower near MG Road Metro Station. The portal's live Geofence listener triggers a critical audio-visual siren alert: *'Critical Geofence Breach: Suspect Rupa Naik pinged near MG Road'*. The Analyst (DA) clicks 'Inspect Trajectory'. The portal immediately switches to the CDR timeline view, mapping the suspect's historical tower coordinates side-by-side with known crime hotspots. The Analyst coordinates with ground patrol units, transmitting the suspect's current location details for immediate interception.

Scenario C: Legislative Crime Trend & Resource Allocation

Operation: The DGP (Policymaker) is preparing the annual budget for police station deployment. The DGP opens the Socio-Demographic charts, analyzing the correlation between poverty indices, police density, and crime rates across Karnataka districts. The DGP notices a critical predictive warning indicating a 20% forecast spike in cybercrimes in Bengaluru Central. The DGP writes a recommendation to redeploy 50 additional cyber-trained officers, updates the findings in the Intelligence Brief editor, and downloads an official classified PDF report to present to the Home Minister.

6. Production Upgrade & Enhancement Plan

To migrate this prototype from the Express-SQLite emulator into a secure, enterprise-grade production environment for the Karnataka State Police, the following five structural upgrades must be completed:

1. Datastore & Semantic Vector Search Migration

The current token-based text search in quickML should be replaced by a production **Catalyst QuickML Vector Index**. Document chunks (case files and suspect records) must be vectorized using the catalyst-z3-embedding-v2 model. This allows semantic search where queries like *'suspect broke lock of jewelry shop'* automatically retrieve cases concerning *'housebreaking by night to steal gold ornament'*, even if the keywords do not match exactly.

2. Live CCTNS & ICJS Integration (ETL Pipelines)

The simulated CCTNS connection must be replaced with official government API adapters. We must deploy hourly **Catalyst Crons** or **Catalyst Circuits** state machines. These workers will fetch new data from the state CCTNS (relational records) and ICJS (court and prison records), transform the records, validate the schemas, write them to the Catalyst Datastore, and update the vector index automatically.

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3. Government-Grade Access Controls & SSO

The simulated role switcher must be deleted. User authentication must route through **Catalyst Authentication** integrated with KSP's central Active Directory or the state government Single Sign-On (SSO) gateway. Database queries must be secured using physical **Row-Level Security (RLS)** policies in the Catalyst Datastore: SIs are restricted to reading cases from their designated police station; DAs are restricted to their district; ACPs and DGPs have statewide visibility. All API routes must use JWT verification.

4. Emulated SDK replacement with Zoho Catalyst Production APIs

The mock handlers inside the shared Catalyst SDK must be rewritten to interface with live APIs:

- **Zia Translation**: Replace the local key-value translation dictionaries in functions/translate with the live *Catalyst Zia Translator*, which supports real-time, bidirectional Kannada-to-English translation.
- **Zia Voice STT/TTS**: Route voice recordings in functions/voice to the *Catalyst Zia Speech-to-Text* API to support diverse localized accents.
- **Zia AutoML**: Train and deploy the recidivism model using *Catalyst Zia AutoML (Tabular)*. The model should be retrained on 100,000+ historic KSP arrest records to generate statistically valid probabilities.
- **SmartBrowz PDF**: Connect the pdf-export function directly to the production *Catalyst SmartBrowz* serverless instance, resolving HTML structures into PDF binary streams.

5. Deployment & Infrastructure Security

Frontend assets must be compiled and deployed onto the **Catalyst Web Client Hosting CDN** for fast load times. The backend Express routes must be deployed as standalone **AppSail** services. The backend endpoints must be secured using a **Catalyst API Gateway** with rate limiting, HTTPS enforcement, and strict CORS policies. Cryptographic audit log chains must be mirrored to KSP's central SIEM console for log aggregation and retention.

CONCLUSION: By implementing these production upgrades, the prototype will shift from a localized mock environment to a highly secure, resilient, and intelligent state-wide crime surveillance system complying with national Indian Law Enforcement cybersecurity standards.